Analytical Inter-Voxel Strain Propagator

Complete Derivation and Numerical Tables

Research Notes

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1 Introduction

The Foldy–Lax multiple-scattering equation

$$\mathbf{T} = \mathbf{T}_0(\mathbf{I} - \mathbf{G}_0\mathbf{T}_0)^{-1} \tag{1}$$

relates the full T-matrix $\mathbf{T}$ of a collection of scatterers to the single-site T-matrix $\mathbf{T}_0$ and the inter-site propagator $\mathbf{G}_0$. For cubic voxels on a lattice, the propagator couples volume-averaged displacement and strain degrees of freedom between sites separated by lattice vector $\boldsymbol{R}$.

The inter-site propagator $P_{ijkl}(\boldsymbol{R}, \omega)$ is the volume-averaged Green's tensor second derivative:

$$P_{ijkl}(\boldsymbol{R}, \omega) \;=\; \int_A \int_B \frac{\partial^2 G_{ik}(\boldsymbol{s} - \boldsymbol{s}' + \boldsymbol{R},\, \omega)}{\partial x_j\, \partial x_l}\, d^3s\, d^3s'. \tag{2}$$

In Fourier space:

$$P_{ijkl}(\boldsymbol{R}, \omega) \;=\; \int \frac{d^3k}{(2\pi)^3}\, \left|\tilde{f}_0(\boldsymbol{k})\right|^2 \tilde{\Gamma}_{ijkl}(\boldsymbol{k}, \omega)\, e^{i\boldsymbol{k}\cdot\boldsymbol{R}}, \tag{3}$$

where $\tilde{f}_0(\boldsymbol{k}) = a^3 \operatorname{sinc}(k_1 h) \operatorname{sinc}(k_2 h) \operatorname{sinc}(k_3 h)/(2\sqrt{\pi})$ is the cube form factor ($h = a/2$) and $\tilde{\Gamma}_{ijkl}$ is the strain kernel.

The full 9×9 block propagator couples 3 displacement and 6 Voigt strain DOF:

$$\mathbf{G}_0(\boldsymbol{R}, \omega) \;=\; \begin{pmatrix} G_{ij} & C_{i\alpha} \\ H_{\alpha i} & S_{\alpha\beta} \end{pmatrix}, \tag{4}$$

where G is the 3×3 Green's tensor block, S is the 6×6 Voigt contraction of P_{ijkl}, C is the 3×6 displacement–strain coupling from $\partial\langle G_{ij}\rangle/\partial R_k$, and $H = C^T$ is its transpose.

This document derives the complete analytical computation of all blocks for the 26 nearest neighbours on a cubic lattice, with dynamic corrections through ω^6.

2 The UV Divergence Problem

For the static (Kelvin) Green's tensor, the strain kernel is

$$\tilde{\Gamma}^{\infty}_{ijkl}(\hat{\boldsymbol{k}}) \;=\; -\frac{1}{2\mu}\left[\hat{k}_j\hat{k}_l\, \delta_{ik} + \hat{k}_i\hat{k}_l\, \delta_{jk} - 2\eta\, \hat{k}_i\hat{k}_j\hat{k}_k\hat{k}_l\right], \tag{5}$$

where $\eta = 1/[2(1-\nu)]$ is the Poisson ratio factor. This depends only on the *direction* $\hat{\boldsymbol{k}}$, not on $|\boldsymbol{k}|$. It is $\mathcal{O}(1)$ as $|\boldsymbol{k}| \to \infty$.

The sinc^2 form factor provides decay $|\tilde{f}_0|^2 \sim 1/k^2$ per axis, giving $1/k^6$ total. But the 3D volume element $k^2\, dk$ and the angular measure contribute k^2, leaving the radial integral as $\int dk/k^4 \times k^2 = \int dk/k^2$, which—combined with the $\mathcal{O}(1)$ kernel—makes the 3D integral (3) **UV-divergent**. This was confirmed numerically: the integral does not converge as $K_{\max} \to \infty$.

The resolution is to trade $\hat{\boldsymbol{k}}_j\hat{\boldsymbol{k}}_l = k_jk_l/k^2$ and $\hat{\boldsymbol{k}}_i\hat{\boldsymbol{k}}_j\hat{\boldsymbol{k}}_k\hat{\boldsymbol{k}}_l = k_ik_jk_kk_l/k^4$ for *real-space derivatives* of convergent potentials, as described in the next section.

3 Static Propagator via Potential Derivatives

3.1 Reduction to convergent potentials

The key idea is that k_jk_l/k^2 in Fourier space corresponds to $-\partial^2/\partial R_j\partial R_l$ acting on the $1/k^2$ potential, and similarly $k_ik_jk_kk_l/k^4$ acts on $1/k^4$. Define:

$$A_{jl}(\boldsymbol{R}) \;=\; \int \frac{d^3k}{(2\pi)^3}\, |\tilde{f}_0|^2\, \hat{k}_j\hat{k}_l\, e^{i\boldsymbol{k}\cdot\boldsymbol{R}} \;=\; -\frac{\partial^2}{\partial R_j\, \partial R_l}\left[\frac{\Phi_{00}(\boldsymbol{R})}{4\pi}\right], \tag{6}$$

$$B_{ijkl}(\boldsymbol{R}) \;=\; \int \frac{d^3k}{(2\pi)^3}\, |\tilde{f}_0|^2\, \hat{k}_i\hat{k}_j\hat{k}_k\hat{k}_l\, e^{i\boldsymbol{k}\cdot\boldsymbol{R}} \;=\; -\frac{\partial^4}{\partial R_i\, \partial R_j\, \partial R_k\, \partial R_l}\left[\frac{\Psi_{00}(\boldsymbol{R})}{8\pi}\right], \tag{7}$$

where the **Newton potential** and **biharmonic potential** are tent-weighted integrals:

$$\Phi_{00}(\boldsymbol{R}) \;=\; \int M_{00}(w_1 - R_1)\, M_{00}(w_2 - R_2)\, M_{00}(w_3 - R_3)\, \frac{1}{|\boldsymbol{w}|}\, d^3w, \tag{8}$$

$$\Psi_{00}(\boldsymbol{R}) \;=\; \int M_{00}(w_1 - R_1)\, M_{00}(w_2 - R_2)\, M_{00}(w_3 - R_3)\, |\boldsymbol{w}|\, d^3w, \tag{9}$$

with $M_{00}(d) = \max(0, 1-|d|)$ the tent function (auto-correlation of two box functions of width a). Both integrals converge absolutely.

The static propagator then assembles as:

$$\boxed{P^{(0)}_{ijkl}(\boldsymbol{R}) \;=\; -\frac{1}{2\mu}\big[\delta_{ik}\, A_{jl} + \delta_{jk}\, A_{il} - 2\eta\, B_{ijkl}\big]} \tag{10}$$

with $\eta = 1/[2(1-\nu)]$.

3.2 Delta-function collapse

The derivatives in (6)–(7) act on the tent functions, not on the kernel. The second derivative of the tent is a sum of delta functions:

$$M''_{00}(d) \;=\; -2\,\delta(d) + \delta(d-1) + \delta(d+1). \tag{11}$$

Each delta **collapses one integration dimension** analytically.

Single delta collapse (for A_{jl}). Differentiating Φ_{00} twice with respect to R_j produces deltas in the w_j integration, leaving a **2D tent-weighted master integral**:

$$I_2^A(p,q,c) \;=\; \int_0^1\!\!\int_0^1 \frac{u^p\, v^q}{\sqrt{c^2+u^2+v^2}}\, du\, dv. \tag{12}$$

Double delta collapse (for B_{ijkl}). When two derivatives act on different tent axes, both dimensions collapse, leaving a **1D elementary integral**:

$$I_1(C) \;=\; \int_0^1 (1-t)\,\sqrt{C+t^2}\, dt \;=\; \frac{1}{2}\left[\sqrt{C+1} + C\,\mathrm{arcsinh}\left(\frac{1}{\sqrt{C}}\right)\right] - \frac{1}{3}\Big[(C+1)^{3/2} - C^{3/2}\Big]. \tag{13}$$

Sequential integration. All 2D masters are computed analytically by Mathematica via sequential integration:

$$\text{inner: } \int_0^1 \frac{u^p\, v^q}{\sqrt{c^2+u^2+v^2}}\, dv \;\xrightarrow{\text{Assumptions}}\; \text{outer: } \int_0^1 (\cdots)\, du \;\xrightarrow{\text{Simplify}}\; \text{exact closed form.} \tag{14}$$

Flags: `Assumptions -> u > 0 && c >= 0`, `GenerateConditions -> False` at every step.

3.3 Numerical values

The Laplacian identity $\sum_k B_{ijkk} = A_{ij}$ is verified to 10^{-14} relative precision for all three neighbour types.

Table 1: Static Newton potential derivatives A_{jl} for unit cube.

Component	Face	Edge	Corner
A_{11}	-0.135017	-0.013786	0.0 (exact)
A_{22}	$+0.067509$	-0.013786	0.0 (exact)
A_{33}	$+0.067509$	$+0.027572$	0.0 (exact)
A_{12}	0	-0.045565	-0.016062
A_{13}	0	0	-0.016062
A_{23}	0	0	-0.016062

Table 2: Static biharmonic potential derivatives B_{ijkl} for unit cube.

Component	Face	Edge	Corner
B_{1111}	-0.105182	-0.021889	-0.006255
B_{1122}	-0.014918	$+0.009657$	$+0.003128$
B_{1133}	-0.014918	-0.001554	$+0.003128$
B_{2222}	$+0.062267$	-0.021889	-0.006255
B_{2233}	$+0.020159$	-0.001554	$+0.003128$
B_{3333}	$+0.062267$	$+0.030680$	-0.006255
B_{1112}	0	-0.016530	-0.009266
B_{1233}	0	-0.012505	—
B_{1123}	0	0	$+0.002470$

4 Three Neighbour Types and O_h Symmetry

On a cubic lattice, the 26 nearest neighbours fall into three orbits under the octahedral group O_h:

Type	Canonical $\boldsymbol{R}/a$	Count	Point group	Independent A_{jl}
Face	$(1,0,0)$	6	C_{4v}	$A_{11},\ A_{22}=A_{33}$
Edge	$(1,1,0)$	12	C_{2v}	$A_{11}=A_{22},\ A_{33},\ A_{12}\neq 0$
Corner	$(1,1,1)$	8	C_{3v}	$A_{11}=A_{22}=A_{33},\ A_{12}=A_{13}=A_{23}$

The independent component counts for each block are:

Type	A_{jl}	B_{ijkl}	d^3W/dR^3	dW/dR
Face (C_{4v})	2	4	2	1
Edge (C_{2v})	3	6	3	2
Corner (C_{3v})	2	4	3	1

The delta-function collapse structure differs for each type:

- **Face**: one tent shifted by a, two centred. Standard 2D masters $I_2^A(p,q,c)$ at $c\in\{0,1,2\}$.
- **Edge**: two tents shifted, one centred. Shifted 2D masters $I^A_{2,\text{shift}}(p,q,c,s)$ with $\sqrt{c^2+(u+s)^2+v^2}$.
- **Corner**: all three tents shifted. Double-shifted 2D masters. But C_{3v} symmetry reduces the count drastically.

O_h rotation to all 26 directions. Any neighbour direction $\boldsymbol{R}$ is related to the canonical direction by a signed-permutation matrix $R\in O_h$. The propagator transforms as:

$$P_{ijkl}(\boldsymbol{R}) \;=\; R_{ia}\,R_{jb}\,R_{kc}\,R_{ld}\,P_{abcd}(\boldsymbol{R}_{\text{can}}). \tag{15}$$

Only 3 canonical directions need be computed; the remaining 23 follow by symmetry.

Corner diagonal vanishing. A notable result is that $A_{11} = A_{22} = A_{33} = 0$ *exactly* for the corner-adjacent case $\boldsymbol{R} = (a, a, a)$. This follows from the identity $B_{1111} = -2B_{1122}$ and the Laplacian relation $B_{1111} + B_{1122} + B_{1133} = A_{11}$.

5 Dynamic Corrections: Power Series in ω^2

5.1 Expansion of the dynamic Green's tensor

The full frequency-dependent Green's tensor in Fourier space is

$$\tilde{G}_{ij}(\boldsymbol{k}, \omega) \;=\; \frac{\hat{k}_i \hat{k}_j}{\rho(\omega^2 - c_p^2 k^2)} \;+\; \frac{\delta_{ij} - \hat{k}_i \hat{k}_j}{\rho(\omega^2 - c_s^2 k^2)}. \tag{16}$$

Expanding for large k ($\omega^2/(c^2 k^2) \ll 1$):

$$\frac{1}{\omega^2 - c^2 k^2} \;=\; -\frac{1}{c^2 k^2} \sum_{n=0}^{\infty} \left(\frac{\omega^2}{c^2 k^2} \right)^n \;=\; -\sum_{n=0}^{\infty} \frac{\omega^{2n}}{c^{2n+2}\, k^{2n+2}}. \tag{17}$$

The strain kernel has the expansion

$$\boxed{\tilde{\Gamma}_{ijkl}(\boldsymbol{k}, \omega) \;=\; \tilde{\Gamma}^{\infty}_{ijkl}(\hat{\boldsymbol{k}}) \;+\; \sum_{n=1}^{\infty} \omega^{2n}\, \frac{\tilde{\Gamma}^{(n)}_{ijkl}(\hat{\boldsymbol{k}})}{k^{2n}}} \tag{18}$$

and therefore

$$P_{ijkl}(\boldsymbol{R}, \omega) \;=\; \sum_{n=0}^{\infty} \omega^{2n}\, P^{(n)}_{ijkl}(\boldsymbol{R}). \tag{19}$$

5.2 The potential hierarchy

Each order n reduces to derivatives of **higher-order potentials**:

$$W_m(\boldsymbol{R}) \;=\; \int M_{00}(w_1 - R_1)\, M_{00}(w_2 - R_2)\, M_{00}(w_3 - R_3)\, |\boldsymbol{w}|^{2m-1}\, d^3 w, \tag{20}$$

connected by the Fourier hierarchy

$$\int \frac{d^3 k}{(2\pi)^3}\, \frac{|\tilde{f}_0|^2}{k^{2m}}\, e^{i\boldsymbol{k}\cdot\boldsymbol{R}} \;=\; c_m\, W_m(\boldsymbol{R}). \tag{21}$$

m	Kernel	Potential	Role	Notation
1	$1/\lvert\boldsymbol{w}\rvert$	Φ (Newton)	Static: A_{jl}	W_1
2	$\lvert\boldsymbol{w}\rvert$	Ψ (biharmonic)	Static: B_{ijkl}; Dyn n=1	W_2
3	$\lvert\boldsymbol{w}\rvert^3$	X (triharmonic)	Dyn n=1, 2	W_3
4	$\lvert\boldsymbol{w}\rvert^5$	Ω (pentaharmonic)	Dyn n=2, 3	W_4
5	$\lvert\boldsymbol{w}\rvert^7$	H (heptaharmonic)	Dyn n=3, 4	W_5

5.3 Laplacian chain

The key relation connecting adjacent potentials is

$$\nabla^2 |\boldsymbol{w}|^{2m+1} \;=\; (2m+1)(2m+2)\, |\boldsymbol{w}|^{2m-1}, \tag{22}$$

giving the **divisors**:

m	Relation	Divisor	Chain
1	$\nabla^2\lvert\boldsymbol{w}\rvert = 2/\lvert\boldsymbol{w}\rvert$	2	$\Psi \to \Phi$
2	$\nabla^2\lvert\boldsymbol{w}\rvert^3 = 12\,\lvert\boldsymbol{w}\rvert$	12	$X \to \Psi$
3	$\nabla^2\lvert\boldsymbol{w}\rvert^5 = 30\,\lvert\boldsymbol{w}\rvert^3$	30	$\Omega \to X$
4	$\nabla^2\lvert\boldsymbol{w}\rvert^7 = 56\,\lvert\boldsymbol{w}\rvert^5$	56	$H \to \Omega$

These imply the Laplacian identities for derivatives:

$$\sum_j \frac{\partial^2 W_{m+1}}{\partial R_j^2} \;=\; (2m+1)(2m+2)\,W_m, \qquad \sum_j B^{(m)}_{ijkk} \;=\; A^{(m)}_{ij}. \tag{23}$$

5.4 Normalisation constants

The normalisation constants c_m relating Fourier integrals to real-space potentials are:

Constant	Value	Use
$c_{A,1} = 1/(8\pi)$	≈ 0.03979	$A^{(1)}$ from $\partial^2\Psi/\partial R^2$
$c_{B,1} = 1/(96\pi)$	≈ 0.003316	$B^{(1)}$ from $\partial^4 X/\partial R^4$
$c_{A,2} = -1/(96\pi)$	≈ -0.003316	$A^{(2)}$ from $\partial^2 X/\partial R^2$
$c_{B,2} = -1/(2880\pi)$	≈ -0.0001105	$B^{(2)}$ from $\partial^4\Omega/\partial R^4$
$c_{A,3} = 1/(2880\pi)$	≈ 0.0001105	$A^{(3)}$ from $\partial^2\Omega/\partial R^2$
$c_{B,3} = 1/(161280\pi)$	$\approx 1.969 \times 10^{-6}$	$B^{(3)}$ from $\partial^4 H/\partial R^4$
$c_{A,4} = -1/(161280\pi)$	$\approx -1.969 \times 10^{-6}$	$A^{(4)}$ from $\partial^2 H/\partial R^2$

Note that $c_{B,n} = c_{A,n}/D_n$ and $c_{A,n+1} = -c_{B,n}$ where D_n is the divisor at order n.

5.5 Deviatoric suppression

At each dynamic order n, the deviatoric (B-tensor) contribution is suppressed by the factor

$$\boxed{\eta_n \;=\; 1 - \left(\frac{c_s}{c_p}\right)^{2n+2}} \tag{24}$$

which approaches unity for large velocity contrast. The static case corresponds to $\eta_0 = 1 - (c_s/c_p)^2 = 1/(2(1-\nu))$.

5.6 Convergence

The series (19) converges for

$$\frac{\omega^2}{c_s^2 k_{\text{eff}}^2} < 1, \qquad \text{i.e.} \qquad ka \lesssim \pi. \tag{25}$$

In the Rayleigh regime ($ka < 0.3$) only $P^{(0)}$ and $P^{(1)}$ are needed. In the transition regime ($0.3 < ka < 1$), three or four terms suffice.

Remark 1 (Smoothness improves with order). The Newton potential ($m = 1$) has a $1/|\boldsymbol{w}|$ singularity—the hardest integral. All higher potentials ($m \geq 2$) have smooth kernels $|\boldsymbol{w}|^{2m-1}$. Successive dynamic corrections are progressively easier to compute.

6 S Block (Strain–Strain, 6×6)

The strain–strain propagator at each order has the same tensorial structure as the static case:

$$\boxed{P^{(n)}_{ijkl}(\boldsymbol{R}) \;=\; -\frac{1}{2\rho c_s^{2n+2}}\left[\delta_{ik}\,A^{(n)}_{jl} + \delta_{jk}\,A^{(n)}_{il} - 2\eta_n\,B^{(n)}_{ijkl}\right]} \tag{26}$$

where $A^{(n)}$ and $B^{(n)}$ are the normalised potential derivatives at order n:

Order n	$A^{(n)}_{jl}$ from	$B^{(n)}_{ijkl}$ from
0 (static)	$\partial^2\Phi/(4\pi\partial R^2)$	$\partial^4\Psi/(8\pi\partial R^4)$
1 (ω^2)	$\partial^2\Psi/(8\pi\partial R^2)$	$\partial^4 X/(96\pi\partial R^4)$
2 (ω^4)	$-\partial^2 X/(96\pi\partial R^2)$	$-\partial^4\Omega/(2880\pi\partial R^4)$
3 (ω^6)	$\partial^2\Omega/(2880\pi\partial R^2)$	$\partial^4 H/(161280\pi\partial R^4)$

The 6×6 Voigt contraction maps $(ij) \to \alpha$ via the standard ordering: $(0,0) \to 1$, $(1,1) \to 2$, $(2,2) \to 3$, $(1,2) \to 4$, $(0,2) \to 5$, $(0,1) \to 6$. Shear columns carry a factor of $1/2$ (engineering strain convention).

6.1 Order 1 (ω^2): $\Psi \to A$, $X \to B$

Table 3: DYN1 raw $A^{(1)}_{jl}$ values ($\times\, c_{A,1} = 1/(8\pi)$).

Component	Face	Edge	Corner
A_{11}	0.3043	0.3853	0.3859
A_{22}	0.8287	0.3853	0.3859
A_{33}	0.8287	0.6464	0.3859
A_{12}	0	-0.2679	-0.1612

Table 4: DYN1 raw $B^{(1)}_{ijkl}$ values ($\times\, c_{B,1} = 1/(96\pi)$).

Component	Face	Edge	Corner
B_{1111}	1.4133	2.2725	2.5330
B_{1122}	1.1194	1.1695	1.0487
B_{1133}	1.1194	1.1816	1.0487
B_{2222}	6.6139	2.2725	2.5330
B_{2233}	2.2113	1.1816	1.0487
B_{3333}	6.6139	5.3935	2.5330
B_{1112}	0	-1.2918	-0.9196
B_{1233}	0	-0.6309	—
B_{1123}	0	0	-0.0954

Table 5: DYN2 raw $A^{(2)}_{jl}$ values ($\times\, c_{A,2} = -1/(96\pi)$).

Component	Face	Edge	Corner
A_{11}	6.1469	6.7280	7.3115
A_{22}	3.9310	6.7280	7.3115
A_{33}	3.9310	4.9261	7.3115
A_{12}	0	1.8087	1.5543

Table 6: DYN2 raw $B^{(2)}_{ijkl}$ values ($\times\, c_{B,2} = -1/(2880\pi)$).

Component	Face	Edge	Corner
B_{1111}	121.11	126.69	133.00
B_{1122}	31.650	40.522	43.170
B_{1133}	31.650	34.625	43.170
B_{2222}	64.730	126.69	133.00
B_{2233}	21.549	34.625	43.170
B_{3333}	64.730	78.533	133.00
B_{1112}	0	22.887	20.489
B_{1233}	0	8.4875	—
B_{1123}	0	0	5.6510

6.2 Order 2 (ω^4): $X \to A$, $\Omega \to B$

6.3 Order 3 (ω^6): $\Omega \to A$, $H \to B$

7 G Block (Displacement–Displacement, 3×3)

The volume-averaged Green's tensor $\langle G_{ij}\rangle$ follows the same potential-derivative pattern but with second derivatives (not fourth):

$$\boxed{\langle G_{ij}\rangle^{(n)} \;=\; (-1)^n \frac{\omega^{2n}}{D_n!\;\rho c_s^{2n+2}} \left[\delta_{ij}\,\overline{W}_n - \frac{\eta_n}{D_n}\,\frac{\partial^2 W_{n+1}}{\partial R_i\,\partial R_j}\right]} \tag{27}$$

where D_n is the Laplacian divisor and $\overline{W}_n = (1/D_n)\sum_k \partial^2 W_{n+1}/\partial R_k^2$ is the trace (from the Laplacian identity).

Explicit formulae at each order.

- **Static** ($n = 0$): $\langle G_{ij}\rangle^{(0)} = \frac{1}{4\pi\mu}\big[\delta_{ij}\Phi - \frac{1}{4(1-\nu)}\partial^2_{ij}\Psi\big]$, where $\Phi = \frac{1}{2}\sum_k \partial^2_{kk}\Psi$.
- ω^2 ($n = 1$): $\langle G_{ij}\rangle^{(1)} = -\frac{\omega^2}{8\pi\rho c_s^4}\big[\delta_{ij}\bar{\Psi} - \frac{\eta_1}{12}\partial^2_{ij}X\big]$, where $\bar{\Psi} = \frac{1}{12}\sum_k \partial^2_{kk}X$.
- ω^4 ($n = 2$): $\langle G_{ij}\rangle^{(2)} = \frac{\omega^4}{96\pi\rho c_s^6}\big[\delta_{ij}\bar{X} - \frac{\eta_2}{30}\partial^2_{ij}\Omega\big]$, where $\bar{X} = \frac{1}{30}\sum_k \partial^2_{kk}\Omega$.
- ω^6 ($n = 3$): $\langle G_{ij}\rangle^{(3)} = -\frac{\omega^6}{2880\pi\rho c_s^8}\big[\delta_{ij}\bar{\Omega} - \frac{\eta_3}{56}\partial^2_{ij}H\big]$, where $\bar{\Omega} = \frac{1}{56}\sum_k \partial^2_{kk}H$.

The ω^6 correction to the G block requires the A-channel of order 4 (second derivatives of H), but *no* B-tensor (fourth derivatives of the next potential would be order 5).

8 C/H Blocks (Displacement–Strain Coupling, $3 \times 6 + 6 \times 3$)

The displacement–strain coupling arises from the rank-3 tensor

$$dG_{ijk}(\boldsymbol{R}) \;=\; \frac{\partial\langle G_{ij}\rangle}{\partial R_k} \tag{28}$$

Table 7: DYN3 raw $A_{jl}^{(3)}$ values ($\times\, c_{A,3} = 1/(2880\pi)$).

Component	Face	Edge	Corner
A_{11}	35.100	51.671	69.712
A_{22}	13.542	51.671	69.712
A_{33}	13.542	25.500	69.712
A_{12}	0	26.145	30.105

Table 8: DYN3 raw $B_{ijkl}^{(3)}$ values ($\times\, c_{B,3} = 1/(161280\pi)$).

Component	Face	Edge	Corner
B_{1111}	1401.0	1840.7	2302.4
B_{1122}	282.28	651.07	800.71
B_{1133}	282.28	401.74	800.71
B_{2222}	356.75	1840.7	2302.4
B_{2233}	119.31	401.74	800.71
B_{3333}	356.75	624.53	2302.4
B_{1112}	0	635.51	708.32
B_{1233}	0	193.08	—
B_{1123}	0	0	269.26

which requires **third derivatives** of the potentials (one order higher than A_{jl}) and **first derivatives** (one order lower, obtained via the Laplacian identity).

At each dynamic order:

$$\boxed{dG_{ijk}^{(n)} \;=\; (-1)^n \frac{\omega^{2n}}{D_n!\;\rho c_s^{2n+2}} \left[\delta_{ij}\, dW_k - \frac{\eta_n}{D_n} \frac{\partial^3 W_{n+1}}{\partial R_i\, \partial R_j\, \partial R_k}\right]} \tag{29}$$

where the first derivative dW_k is obtained from the Laplacian:

$$dW_k \;=\; \frac{1}{D_n} \sum_j \frac{\partial^3 W_{n+1}}{\partial R_j^2\, \partial R_k}. \tag{30}$$

The H block (6×3) is assembled by Voigt contraction:

$$H_{\alpha i} \;=\; \sum_{jk} V_{\alpha jk}\, dG_{ijk}, \qquad C_{i\alpha} \;=\; \sum_{jk} V_{\alpha jk}\, dG_{jki}, \qquad H = C^T. \tag{31}$$

8.1 Static C/H: $d^3\Psi/dR^3$ and $d\Phi/dR$

8.2 ω^2: d^3X/dR^3 and $d\Psi/dR$

8.3 ω^4: $d^3\Omega/dR^3$ and dX/dR

8.4 ω^6: d^3H/dR^3 and $d\Omega/dR$

9 Validation and Summary

9.1 Laplacian identities

The fundamental check is the Laplacian identity

$$\sum_k B_{ijkk}^{(n)} \;=\; A_{ij}^{(n)} \tag{32}$$

Table 9: DYN4 raw $A_{jl}^{(4)}$ values for G block ω^6 correction ($\times\, c_{A,4} = -1/(161280\pi)$).

Component	Face	Edge	Corner
A_{11}	166.04	336.61	569.49
A_{22}	47.160	336.61	569.49
A_{33}	47.160	128.50	569.49
A_{12}	0	208.61	311.88

Table 10: Biharmonic third derivatives $d^3\Psi/dR_i\, dR_j\, dR_k$ for static C/H.

Component	Face	Edge	Corner
D_{000}	-0.795438	-0.501864	-0.346474
D_{011}	-0.528262	$+0.048829$	-0.021876
D_{022}	-0.528262	-0.262771	-0.021876
D_{012}	—	—	$+0.142589$

at every dynamic order for all three neighbour types. This is verified to 10^{-9}–10^{-12} relative precision.

For the C/H blocks, the analogous identity is

$$\sum_j \frac{\partial^3 W_{m+1}}{\partial R_j^2\, \partial R_k} \;=\; D_m \cdot \frac{\partial W_m}{\partial R_k} \tag{33}$$

with divisors $D_m \in \{2, 12, 30, 56\}$. This is verified to 18+ digits for the static case and 12+ digits for dynamic orders.

9.2 Point-group symmetry

All tensor components respect the appropriate point-group symmetry:

- **Face** (C_{4v}): $A_{22} = A_{33}$, $A_{12} = A_{13} = A_{23} = 0$, $B_{2222} = B_{3333}$, $B_{1122} = B_{1133}$, all mixed $B = 0$.
- **Edge** (C_{2v}): $A_{11} = A_{22}$, $A_{13} = A_{23} = 0$, $B_{1111} = B_{2222}$, $B_{1133} = B_{2233}$, $B_{1112} = B_{1222}$, $B_{1123} = B_{2213} = 0$.
- **Corner** (C_{3v}): full S_3 permutation symmetry. $A_{11} = A_{22} = A_{33} = 0$ (exact), all off-diagonal A equal, $B_{1111} = -2B_{1122}$.

9.3 $H = C^T$ preservation

The identity $H_{\alpha i} = C_{i\alpha}$ (transpose symmetry of the coupling blocks) is preserved at all dynamic orders. This follows from the symmetry of dG_{ijk} in its first two indices (inherited from $G_{ij} = G_{ji}$).

9.4 Inversion symmetry

Under $\boldsymbol{R} \to -\boldsymbol{R}$:

- $G_{ij}(-\boldsymbol{R}) = G_{ij}(\boldsymbol{R})$ — even (second derivatives of even potentials).
- $C_{i\alpha}(-\boldsymbol{R}) = -C_{i\alpha}(\boldsymbol{R})$ — odd (first/third derivatives pick up a sign).
- $S_{\alpha\beta}(-\boldsymbol{R}) = S_{\alpha\beta}(\boldsymbol{R})$ — even (fourth derivatives of even potentials).

Table 11: Newton first derivatives $d\Phi/dR_k$ (from Laplacian: $d\Phi_k = \frac{1}{2}\sum_j D^{\Psi}_{jjk}$).

Component	Face	Edge	Corner
$d\Phi_0$	-0.925981	-0.357903	-0.195113
$d\Phi_1$	0	-0.357903	-0.195113
$d\Phi_2$	0	0	-0.195113

Table 12: Triharmonic third derivatives $d^3X/dR_i\,dR_j\,dR_k$ (ω^2 C/H correction).

Component	Face	Edge	Corner
D_{000}	$+5.5907$	$+4.7979$	$+4.2544$
D_{011}	$+2.2218$	$+1.1778$	$+1.1463$
D_{022}	$+2.2218$	$+1.8051$	$+1.1463$
D_{012}	—	—	-0.4109

9.5 Convergence

Each successive dynamic order is smaller than the previous one for $ka < 1$:

$$\frac{|P^{(n+1)}|}{|P^{(n)}|} \sim \left(\frac{ka}{\pi}\right)^2 \ll 1 \quad \text{for} \quad ka < 1. \tag{34}$$

9.6 Dynamic coverage summary

Block	Size	ω^0	ω^2	ω^4	ω^6
S (strain–strain)	6×6	✓	✓	✓	✓
G (disp.–disp.)	3×3	✓	✓	✓	✓
C (disp.–strain)	3×6	✓	✓	✓	✓
H (strain–disp.)	6×3	✓	✓	✓	✓

All four blocks of the 9×9 propagator are complete through ω^6, for all 26 nearest neighbours. The total number of precomputed analytical constants is approximately 200.

9.7 Potential hierarchy

The potentials used at each block and order are:

	Φ $(1/\rho)$	Ψ (ρ)	X (ρ^3)	Ω (ρ^5)	H (ρ^7)
S: $\partial^2/\partial R^2$ (A)	ω^0	ω^2	ω^4	ω^6	—
S: $\partial^4/\partial R^4$ (B)	—	ω^0	ω^2	ω^4	ω^6
G: $\partial^2/\partial R^2$	—	ω^0	ω^2	ω^4	ω^6
C/H: $\partial^3/\partial R^3$	—	ω^0	ω^2	ω^4	ω^6
C/H: $\partial/\partial R$	ω^0	ω^2	ω^4	ω^6	—

No numerical quadrature is needed at any stage. Python consumes the precomputed constants for the Foldy–Lax solver.

Table 13: Biharmonic first derivatives $d\Psi/dR_k$ ($d\Psi_k = \frac{1}{12}\sum_j D^X_{jjk}$).

Component	Face	Edge	Corner
$d\Psi_0$	+0.836186	+0.648407	+0.545583
$d\Psi_1$	0	+0.648407	+0.545583
$d\Psi_2$	0	0	+0.545583

Table 14: Pentaharmonic third derivatives $d^3\Omega/dR_i\,dR_j\,dR_k$ (ω^4 C/H correction).

Component	Face	Edge	Corner
D_{000}	+74.703	+86.935	+97.745
D_{011}	+21.532	+34.628	+37.518
D_{022}	+21.532	+26.153	+37.518
D_{012}	—	—	+7.4211

Table 15: Triharmonic first derivatives dX/dR_k ($dX_k = \frac{1}{30}\sum_j D^\Omega_{jjk}$).

Component	Face	Edge	Corner
dX_0	+3.9256	+4.9239	+5.7594
dX_1	0	+4.9239	+5.7594
dX_2	0	0	+5.7594

Table 16: Heptaharmonic third derivatives $d^3H/dR_i\,dR_j\,dR_k$ (ω^6 C/H correction).

Component	Face	Edge	Corner
D_{000}	+520.37	+818.18	+1154.5
D_{011}	+119.39	+401.75	+531.43
D_{022}	+119.39	+208.56	+531.43
D_{012}	—	—	+219.48

Table 17: Pentaharmonic first derivatives $d\Omega/dR_k$ ($d\Omega_k = \frac{1}{56}\sum_j D^H_{jjk}$).

Component	Face	Edge	Corner
$d\Omega_0$	+13.556	+25.509	+39.595
$d\Omega_1$	0	+25.509	+39.595
$d\Omega_2$	0	0	+39.595